



INSTALLATION INSTRUCTIONS

V1.2 November 2024

Firmware 006_001_23



SET UP

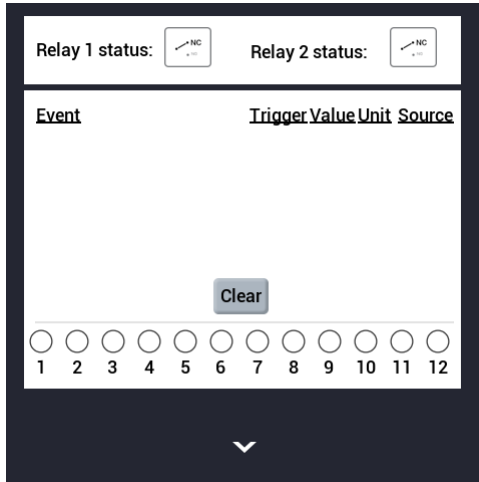
Refer to the Instructions for the required Wiring configuration

Do not connect the relay contacts until set up has been completed.

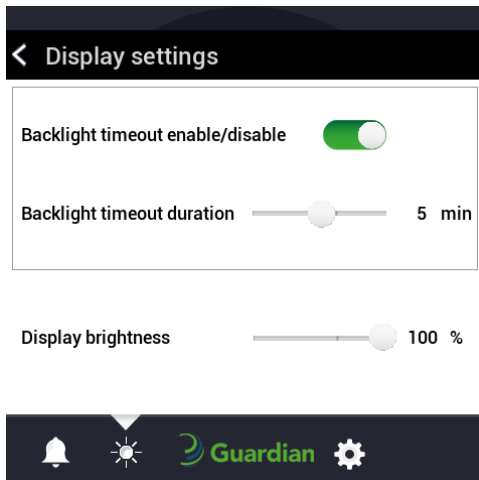
Once the unit has been energised, parameters can be configured to specific values or left as default settings.



MAIN SCREEN



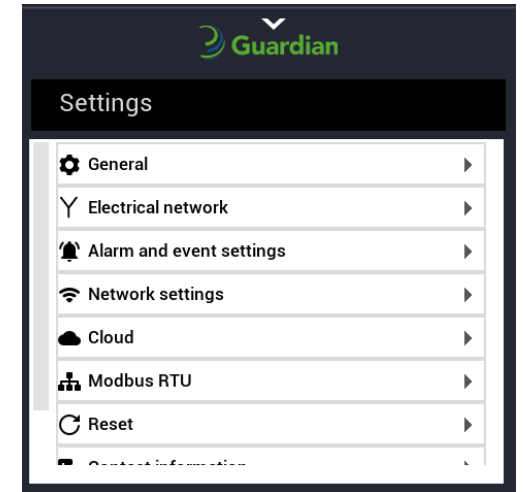
Events screen



Display settings



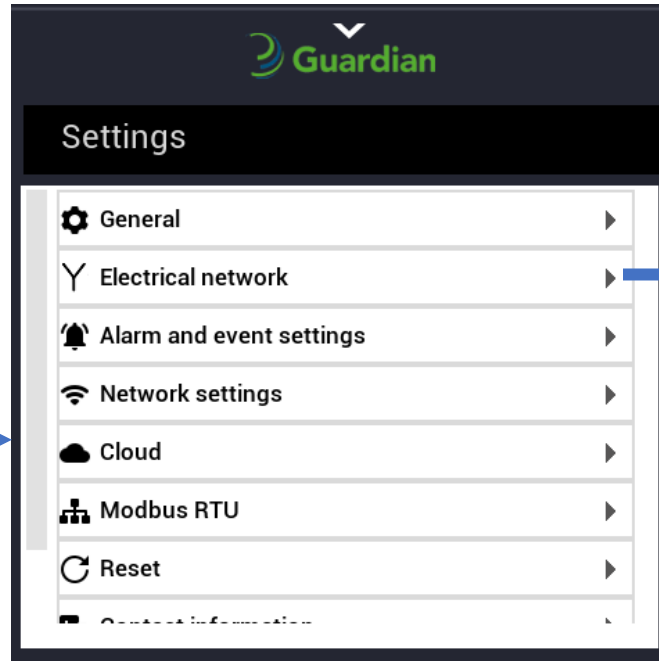
Screen lock
(Press and hold for 3 seconds)



Settings



SETTINGS SCREEN



Electrical network settings

Electrical network configure

1P2W	1P2W+E	1P3W	3P4W	3P3W
☐	☐	☐	☒	☐

Nominal network voltage: 400 Volts

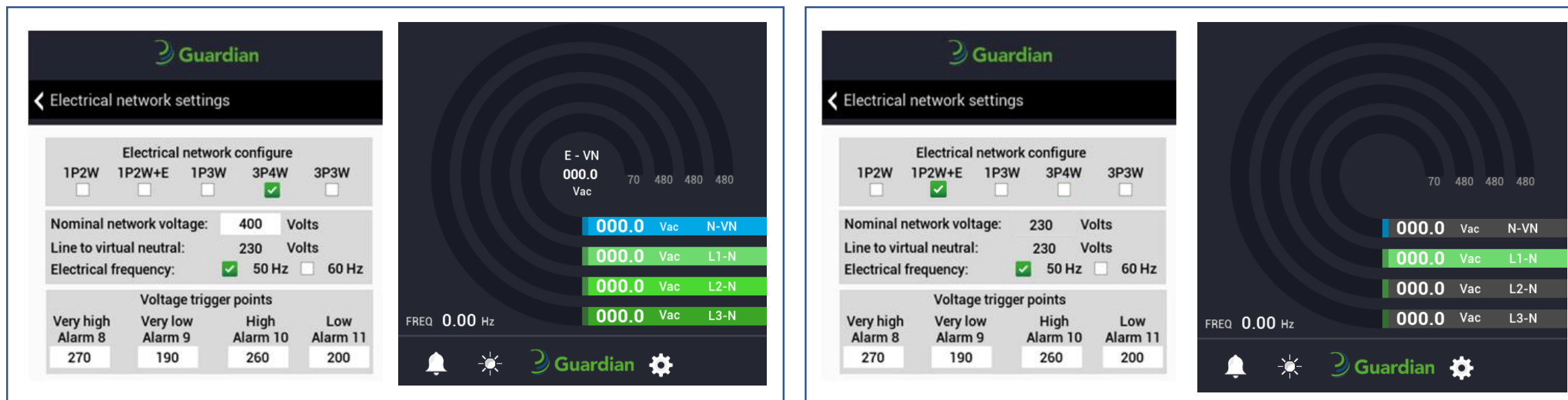
Line to virtual neutral: 230 Volts

Electrical frequency: ☒ 50 Hz ☐ 60 Hz

Voltage trigger points

Very high Alarm 8	Very low Alarm 9	High Alarm 10	Low Alarm 11
270	190	260	200

ELECTRICAL NETWORK SETTINGS



- Select system electrical network choice.
- Selecting an electrical network configuration will change the main screen display information.
- 1P2W will show L1 to Neutral only.
- 1P2W + E will show L1 to Neutral and Earth to Neutral.
- 1P3W will show L1 to N, L2 to N, N to VN and E to VN.
- 3P4W will show L1 to N, L2 to N, L3 to N N to VN and Earth to VN.
- 3P3W will show L1 to VN, L2 to VN, L3 to VN and Earth to VN.



ELECTRICAL NETWORK SETTINGS



Electrical network settings

Electrical network configure

1P2W	1P2W+E	1P3W	3P4W	3P3W
☐	☐	☐	☒	☐

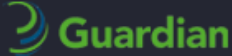
Nominal network voltage: 400 Volts

Line to virtual neutral: 230 Volts

Electrical frequency: ☒ 50 Hz ☐ 60 Hz

Voltage trigger points

Very high Alarm 8	Very low Alarm 9	High Alarm 10	Low Alarm 11
270	190	260	200



Electrical network settings

1PW2 ☐ 3P3W ☐

Nominal network voltage: 400 Volts

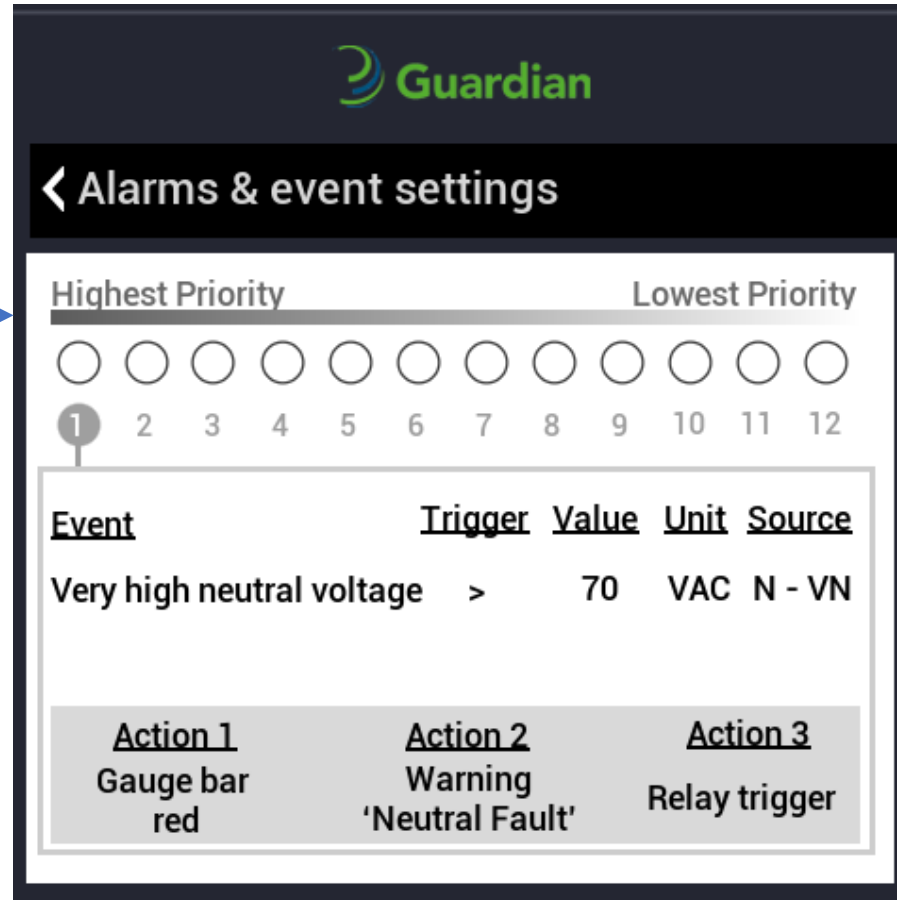
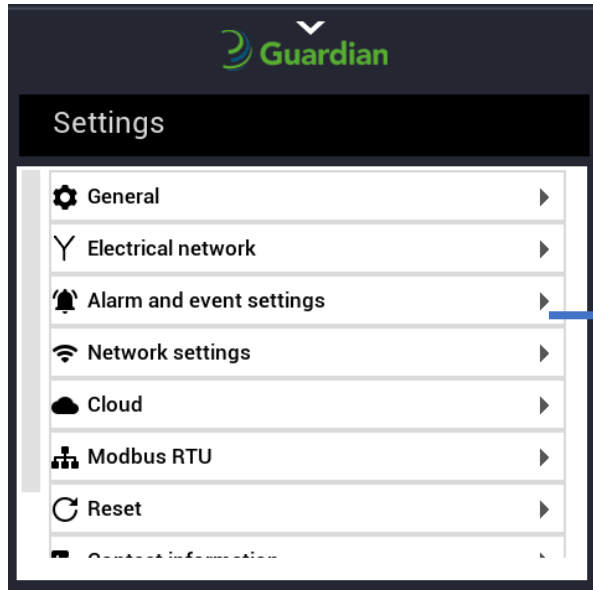
Nominal network frequency: 50 Hz

Nominal network phase: 3

Cancel 0 Del Enter

- Select the nominal network voltage to edit the value.
- If number > 480 VAC “Out of range” will be displayed.
- Press cancel, to not make a change.
- Press enter to confirm change.
- The nominal line voltage will then be updated.
- Events related to the nominal network voltage, will update.

ALARMS AND EVENTS SETTINGS



- 13 event settings.
- Select the number to access the event of interest.
- The event with highest priority will trigger over lower priority.

CONFIGURABLE EVENTS

Event 2 - High neutral voltage (20v) Range from 1-65v N-VN No Relay/Relay Trigger

Event 4 - High earth voltage (20v) Range from 1-60v N-VN No Relay/Relay Trigger

Event 8 - Very high line voltage (260v) Range from 90v-270v No Relay/Relay Trigger

Event 9 - Very low line voltage (190v) Range from 90v-270v No Relay/Relay Trigger

Event 10 - High line voltage (250v) Range from 100v-260v Alert only

Event 11 - Low line voltage (200v) Range from 100v-260v Alert only

Event 12 - Frequency out of range (0.5hz) Range from 0.5-19.9hz No Relay/Relay Trigger

Please note, some events may not appear depending on your supply electrical network.



ALARMS AND EVENTS SETTINGS

Guardian

< Alarms & event settings

Highest Priority Lowest Priority

1 2 3 4 5 6 7 8 9 10 11 12

Event	Trigger	Value	Unit	Source
Very high line voltage	>	260	VAC	L1 - N L2 - N L3 - N
Line to neutral voltage				

Action 1 Gauge bar red

Action 2 Warning: 'Very High Line Voltage'

Action 3 Relay Trigger

Guardian

< Alarms & event settings

Highest Priority Lowest Priority

1 2 3 4 5 6 7 8 9 10 11 12

Event	Trigger	Value	Unit	Source
Very high line voltage	>	260	VAC	L1 - N L2 - N L3 - N
Line to neutral voltage				

Action 1 Gauge bar red

Relay Trigger OFF ON

Action 3 Relay Trigger

- Select the box value “Relay Trigger”. To toggle the relay ON or OFF
- Press the “X” button to close the keypad.
- Events 8, 9, 10 and 11 are effected by the electrical network nominal voltage.

To Amend the voltage value revert back to the electrical network settings.

When all the configurable values have been checked or amended from default, connect the relays.



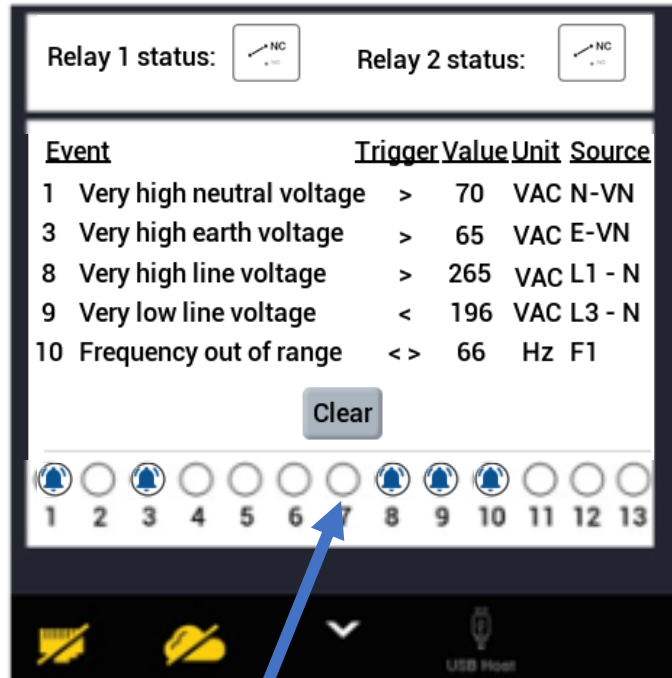
EVENTS EXPLANATION

- Event 1 - Very high neutral voltage (Relay Trigger)
- Event 2 - High neutral voltage (Range from 1-65v N-VN No Relay/Relay Trigger)
- Event 3 - Very high earth voltage (No Relay/Relay Trigger)
- Event 4 - High earth voltage (Range from 1-60v N-VN No Relay Trigger)
- Event 5 - Voltage overline range (Relay Trigger)
- Event 6 - Loss of Phase (No Relay/Relay Trigger)
- Event 7 - Frequency out of range (No Relay/Relay Trigger)
- Event 8 - Very high line voltage (Range from 90v-270v, No Relay/Relay Trigger)
- Event 9 - Very low line voltage (Range from 90v-270v, No Relay/Relay Trigger)
- Event 10 - High line voltage (Range from 100v-260v, Alert only)
- Event 11 - Low line voltage (Range from 100v-260v, Alert only)
- Event 12 - Frequency out of range (Range Margin from 0.5 - 19.9hz, No Relay/Relay Trigger)
- Event 13 – Phase voltage imbalance (Imbalance of 3% to any other phase, Alert only)

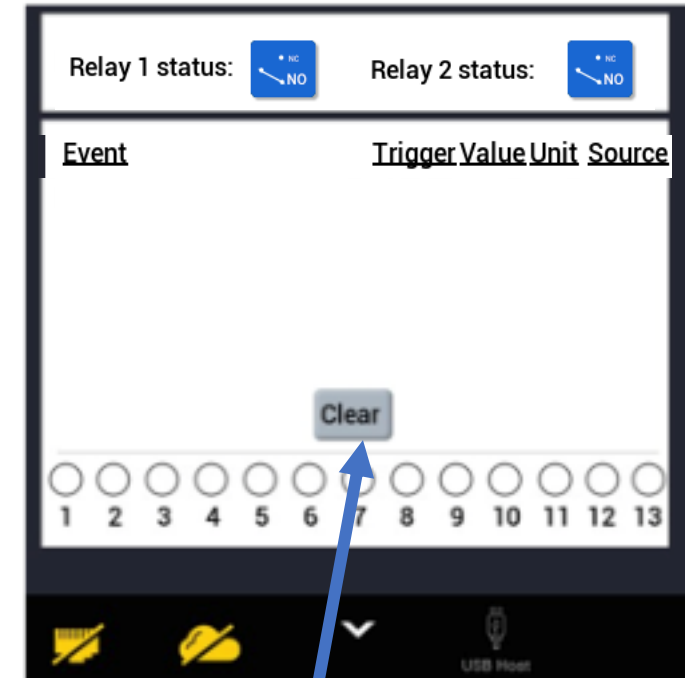
Please note, some events may not appear depending on your supply electrical network.



EVENTS SCREEN EXAMPLE



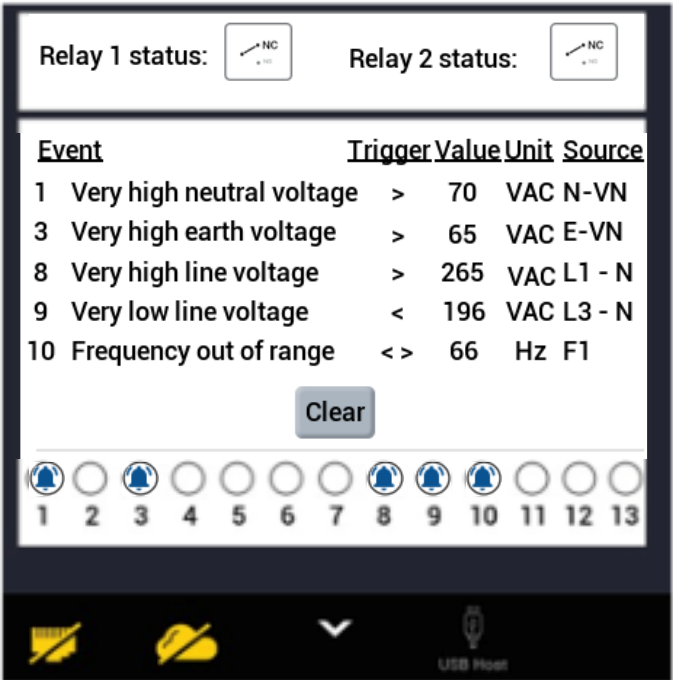
Events that have been triggered



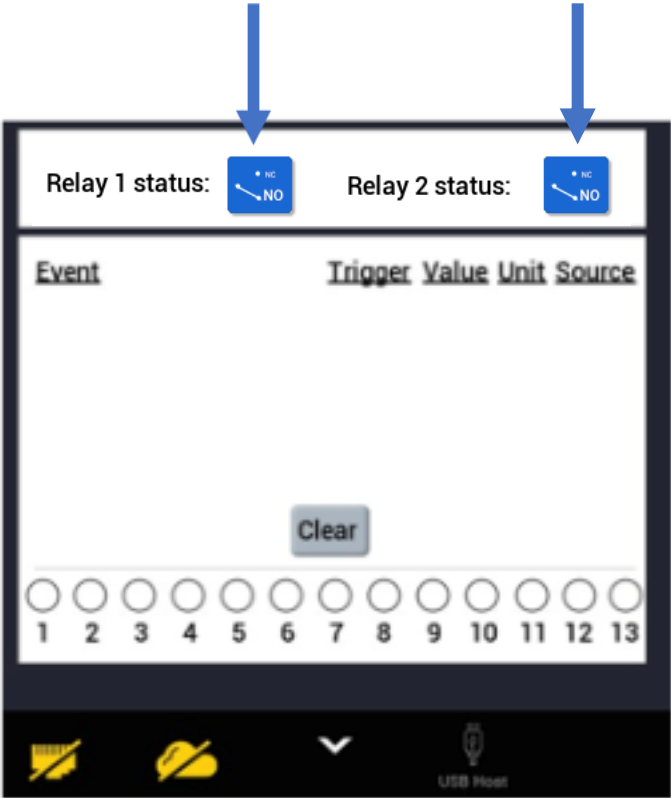
Events can be cleared here

FAULT CONDITIONS

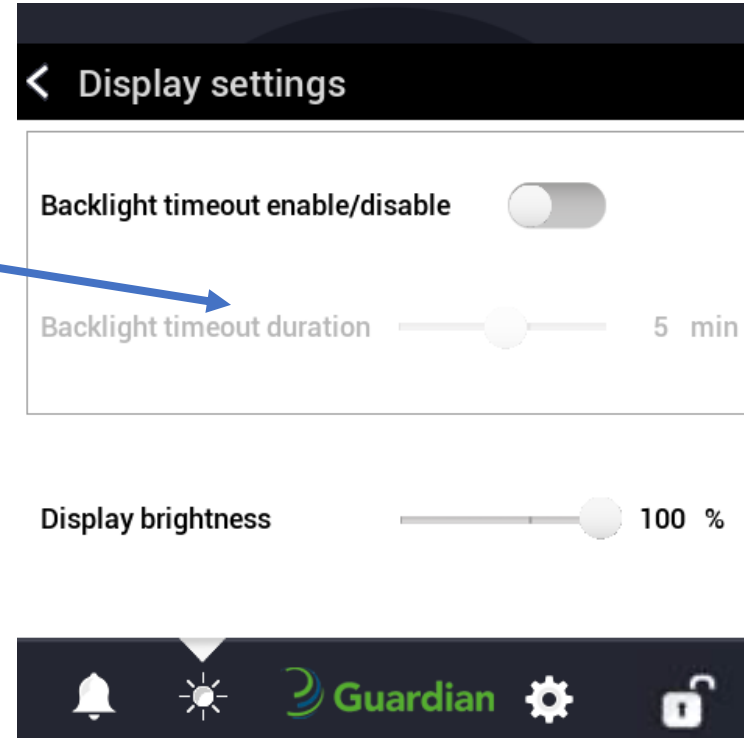
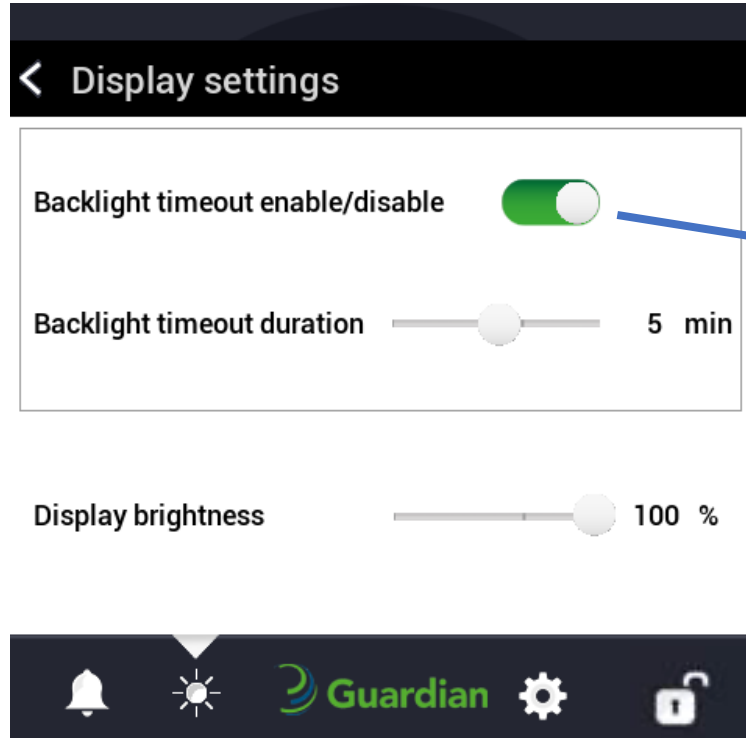
When a fault has been detected the Guardian will Switch both relays to the NC state, in turn switching The connected contactor or UV Device off.



Once a detected fault has cleared, the Guardian will wait a further 3 minutes before changing the Relay status to NO state.



DISPLAY SETTINGS



- Backlight timeout switch will enable or disable access to the time duration.
- Drag the scroll bars to change the settings.

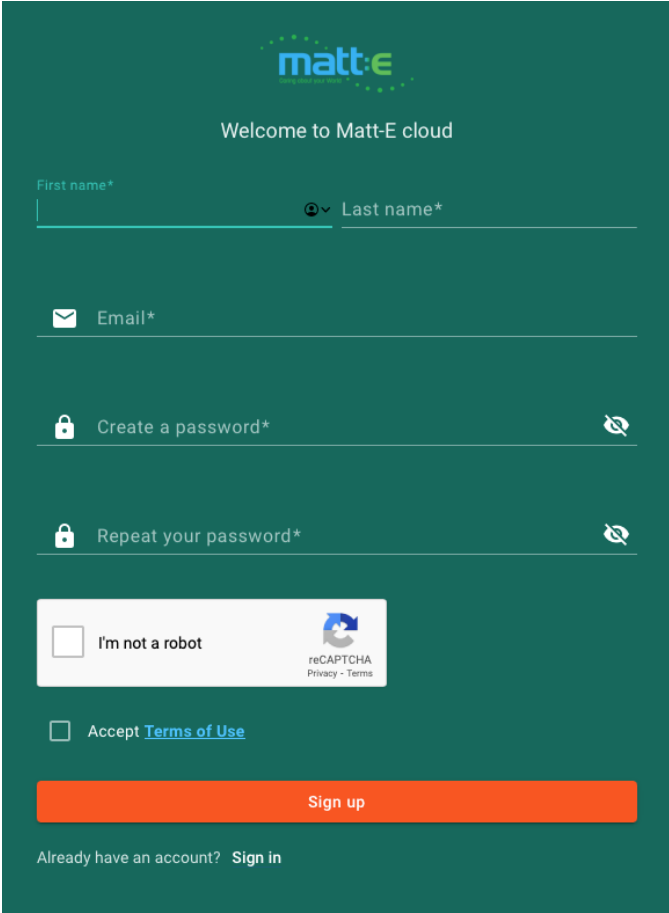
Claiming a device

Create an account on the sign up page of the cloud.

cloud.matt-e.com/login is the URL, click on the sign up button on this page.

Enter your information to create an account in the relative field.

Verify your email address from your chosen email platform.



The image shows a sign-up form for 'matt:e cloud'. The form is set against a dark teal background. At the top, the 'matt:e' logo is displayed in blue and green, followed by the text 'Welcome to Matt-E cloud'. The form fields are as follows: 'First name*' and 'Last name*' (with a dropdown arrow icon), 'Email*' (with an envelope icon), 'Create a password*' (with a lock icon and an eye icon for toggling visibility), and 'Repeat your password*' (with a lock icon and an eye icon). Below these fields is a white box containing a reCAPTCHA 'I'm not a robot' checkbox and a reCAPTCHA logo with links for 'Privacy' and 'Terms'. Underneath is another checkbox for 'Accept Terms of Use', where 'Terms of Use' is a blue hyperlink. A prominent orange 'Sign up' button is located below the checkboxes. At the very bottom, there is a link that says 'Already have an account? Sign in'.



MATT-E CLOUD HUB



DEVICE LIST

Status

Name ↑

Last Update Time

No entities found

Items per page: 10

1 - 0 of 0

|< < > >|

Welcome
to the **Guardian** cloud



Guardian v1

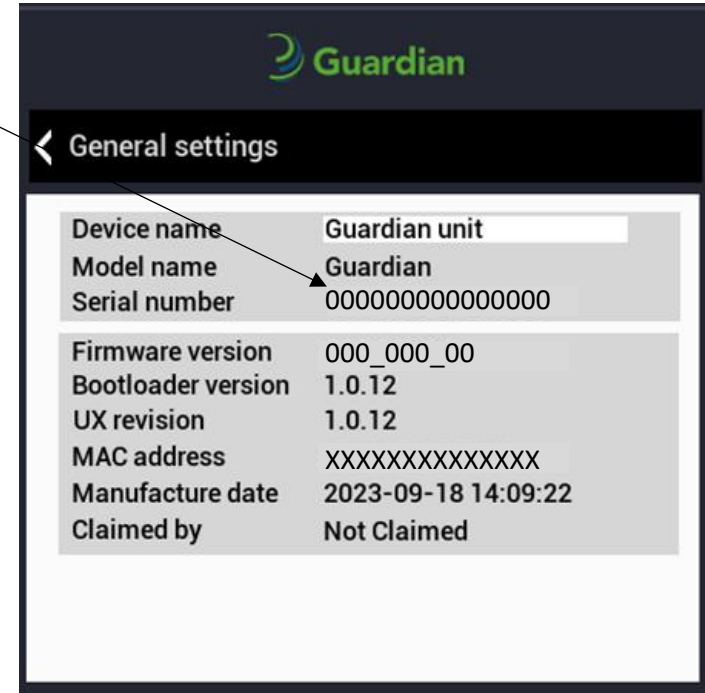
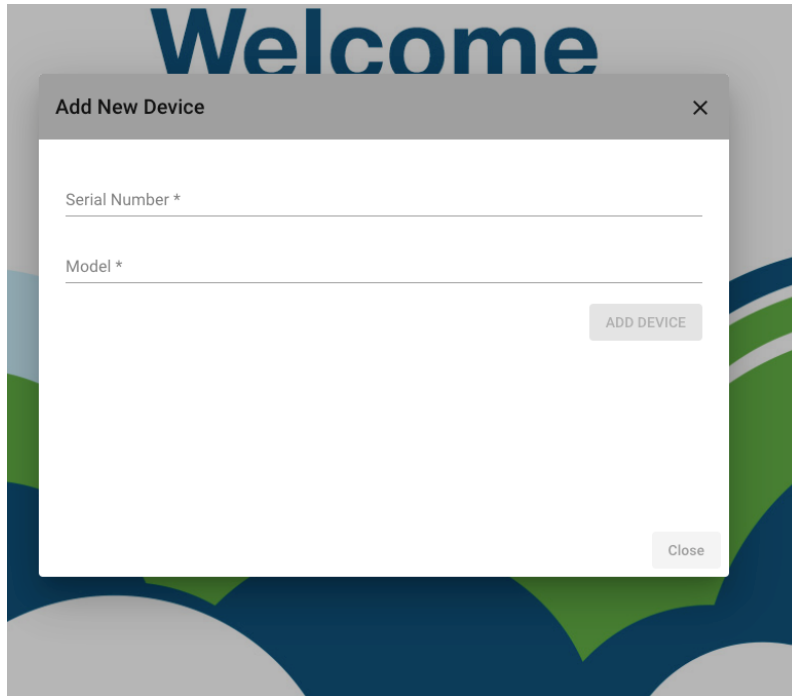


CLAIM THE DEVICE

To claim a device, click on the + icon in the device list. You will be asked to enter a serial number and a model.

The model will always be 'Guardian' (case sensitive)

The serial numbers can be taken from the device (on the device, go to settings > general settings)

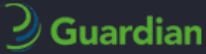


Device name	Guardian unit
Model name	Guardian
Serial number	0000000000000000
Firmware version	000_000_00
Bootloader version	1.0.12
UX revision	1.0.12
MAC address	XXXXXXXXXXXXXX
Manufacture date	2023-09-18 14:09:22
Claimed by	Not Claimed

Click "Add Device"



CLAIMED DEVICE



< General settings

Device name	Guardian unit
Model name	Guardian
Serial number	0000000000000000
Firmware version	006.00.18
Bootloader version	1.0.18
UX revision	1.0.13
MAC address	XXXXXXXXXXXXXX
Manufacture date	2023-09-21 10:44:43
Claimed by	j*****@matt-e.co.uk

Confirmation of a claimed unit



< General settings

Device name	Guardian unit
Model name	Guardian
Serial number	0000000000000000
Firmware version	006.00.18
Bootloader version	1.0.18
UX revision	1.0.13
MAC address	XXXXXXXXXXXXXX
Manufacture date	2023-09-21 10:44:43
Claimed by	No cloud - Connect for info

Confirmation of a claimed unit but
The connection has been lost

VIEW DEVICES

Once logged in the home page lists the connected devices on the left hand side.



DEVICE LIST

status	Enabled	Name ↑	Last Update Time
ALARM	YES	Guardian 1	24/04/2023,11:31:16
ON	YES	Guardian 2	24/04/2023,11:31:04
ON	YES	Guardian 3	24/04/2023,11:31:04

1 - 3 of 3

matt:e

Welcome
to the **Guardian** cloud

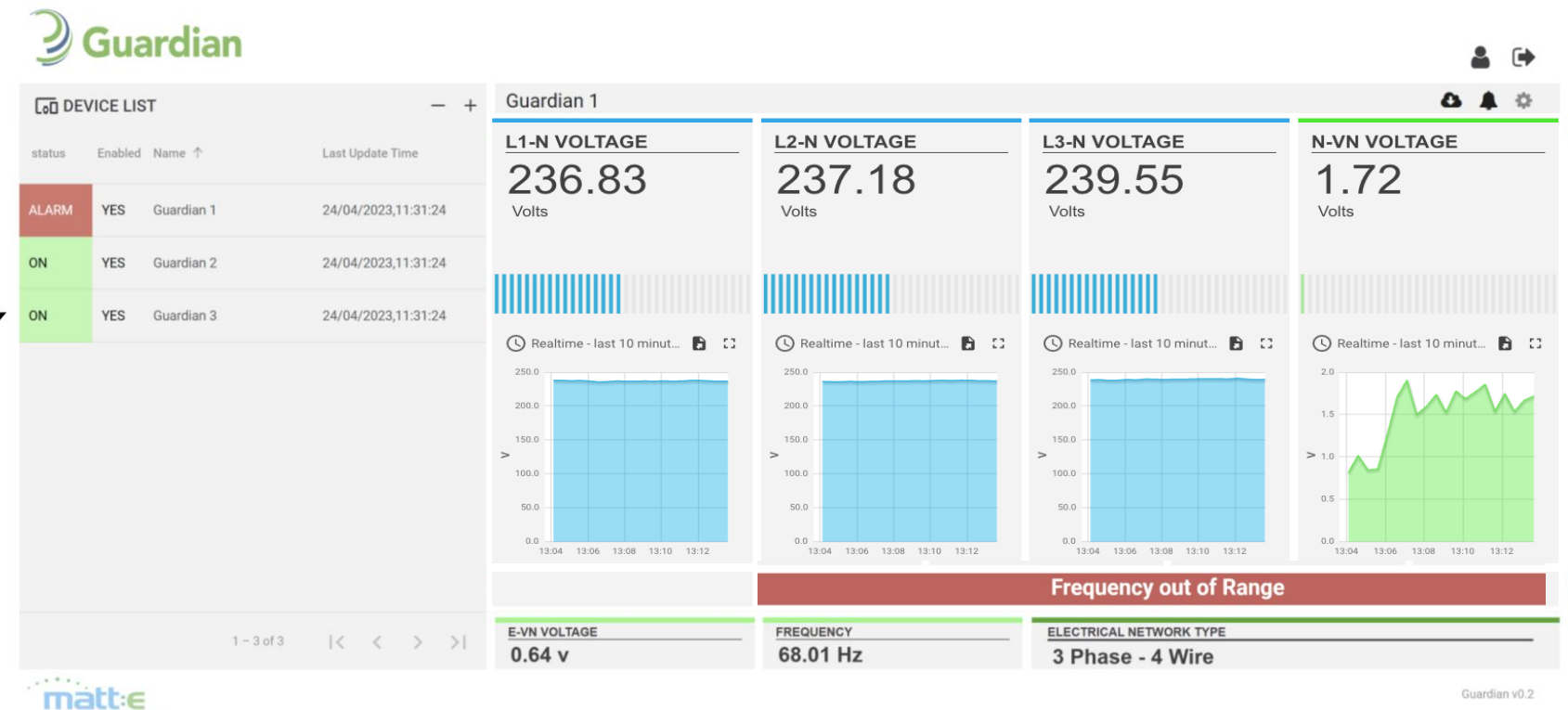


Guardian v0.2

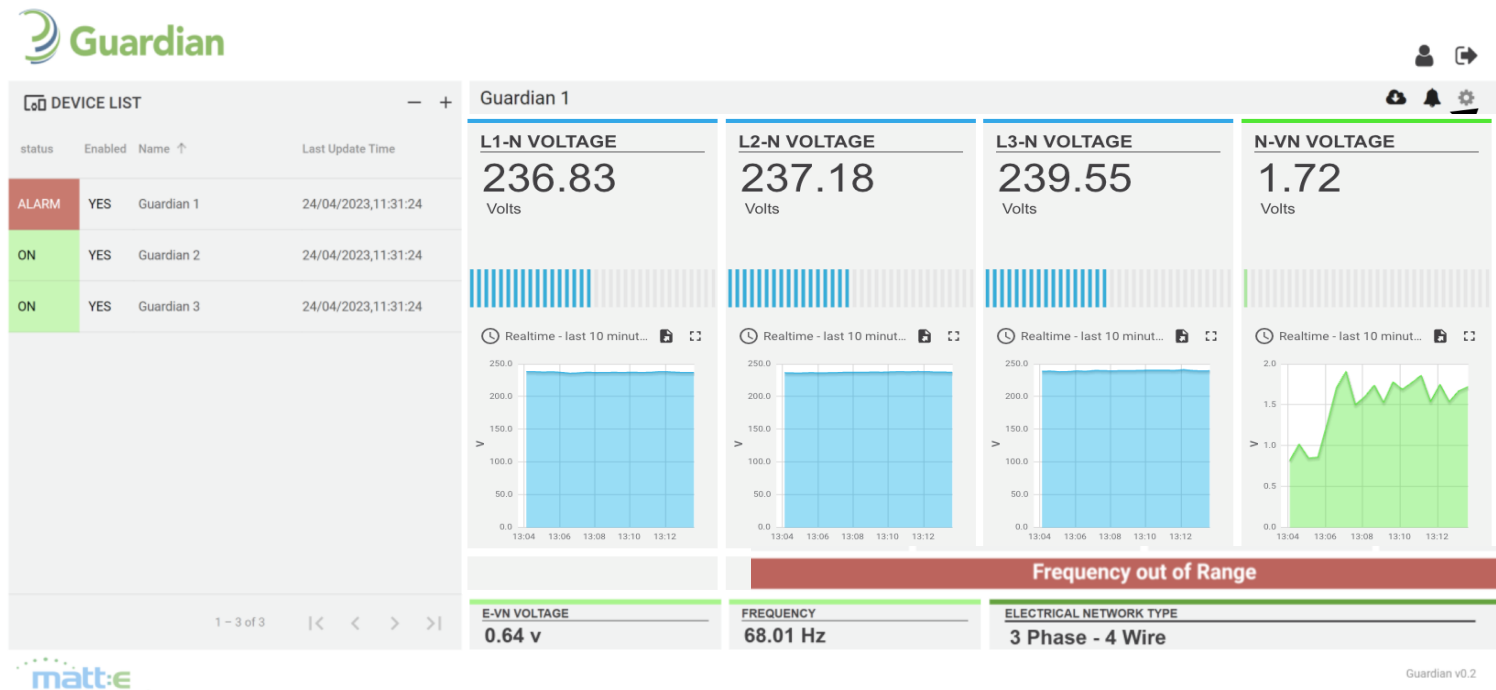


DEVICE HOME SCREEN

Click on one of the connected devices to see the measured values.



EVENTS LOG



Click on the bell icon to show the event log as below.



DOWNLOAD EVENTS



☰ DEVICE LIST

status

Enabled

Name ↑

Last Update Time

ALARM	YES	Guardian 1	24/04/2023,11:31:24
ON	YES	Guardian 2	24/04/2023,11:31:24
ON	YES	Guardian 3	24/04/2023,11:31:24

1 - 3 of 3 |< < > >|

Guardian 1

🔔 EVENT LOGS

🕒 Realtime - last 30 minutes

Priority

source

Status ↑

details

Created time

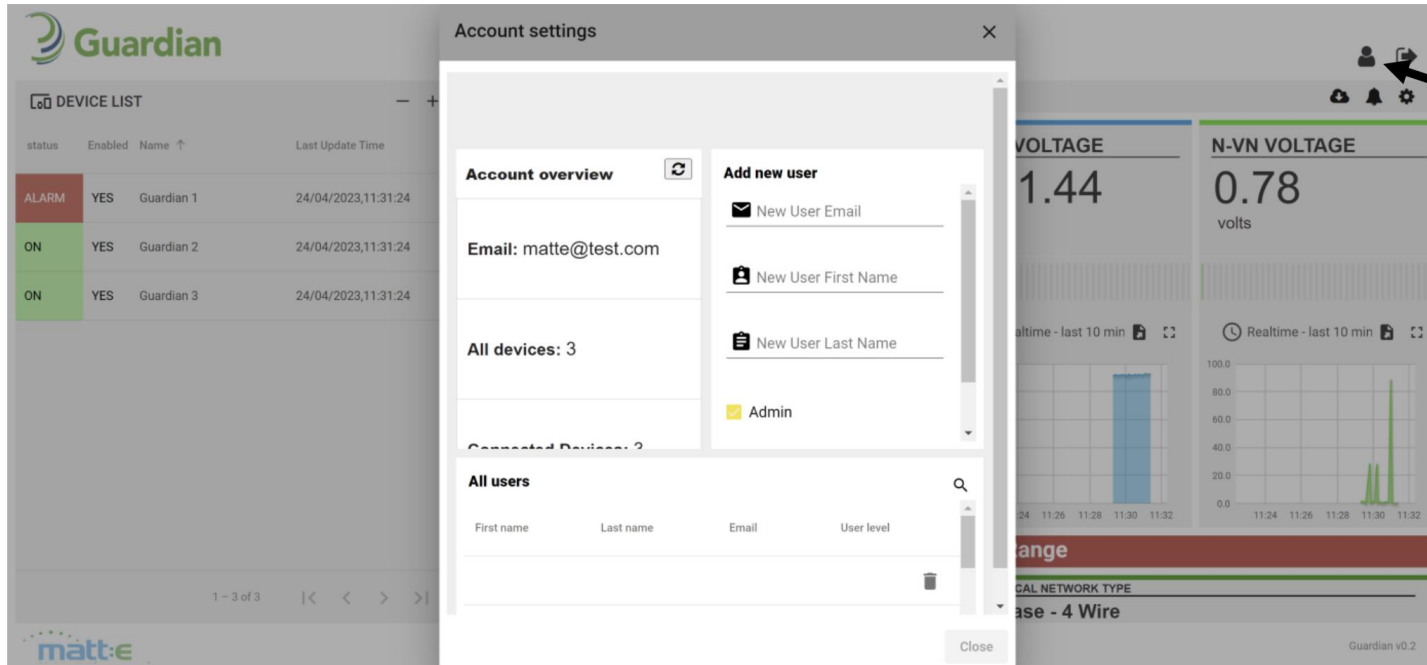
Cleared time

11	Frequency	Triggered	Frequency out of Range	2023-04-24 11:31:24	
9	L1	Cleared	Very Low Line Voltage	2023-04-24 11:31:16	2023-04-24 11:31:24
1	N-VN	Cleared	Very High Neutral Voltage	2023-04-24 11:30:59	2023-04-24 11:31:04
3	E-VN voltage	Cleared	Very High Earth Voltage	2023-04-24 11:30:38	2023-04-24 11:30:40
9	L1	Cleared	Very Low Line Voltage	2023-04-24 11:30:26	2023-04-24 11:30:29
2	N-VN	Cleared	High Neutral Voltage	2023-04-24 11:30:11	2023-04-24 11:30:16
11	Frequency	Cleared	Frequency out of	2023-04-24 11:30:53	2023-04-24 11:30:01

Items per page: 10 | 1 - 10 of 10 |< < > >|

This log can be
downloaded here as
an excel file

SET UP USER DETAILS



The user icon allows you to set a new user. Enter the contact name and email address. You can also select if you want them to be admin or read only email

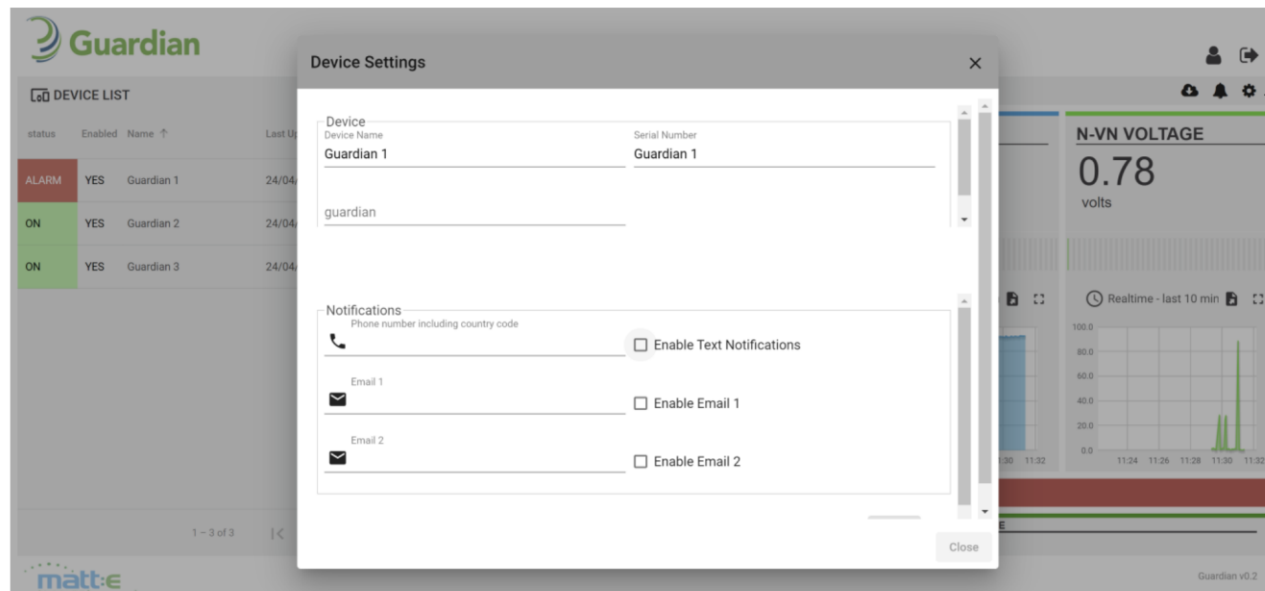


DEVICE NOTIFICATION SET UP

To receive notification of the events via text and/or email enter your details in the Device Settings

Ensure to select the “Enable” box to the right.

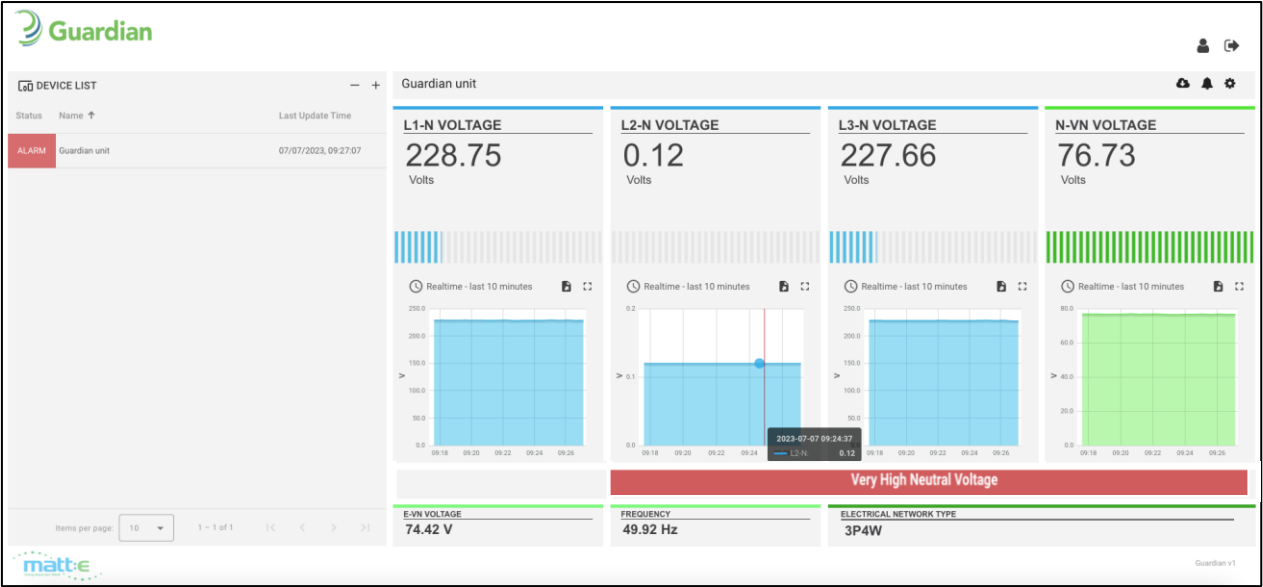
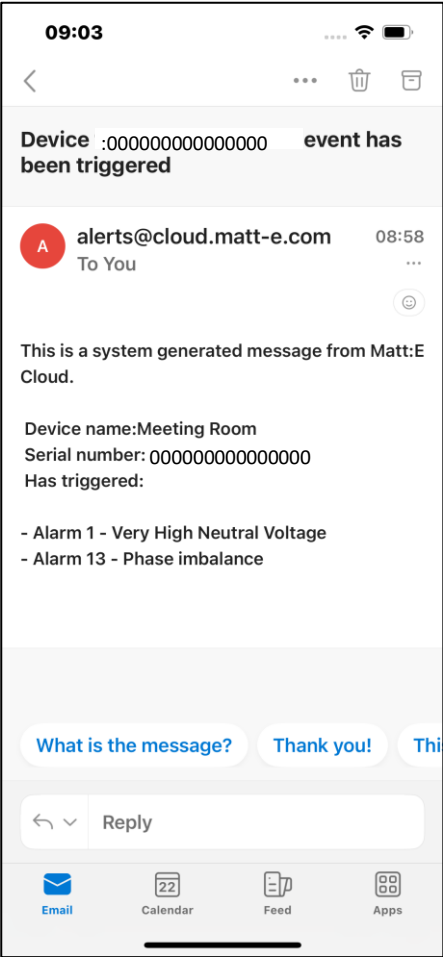
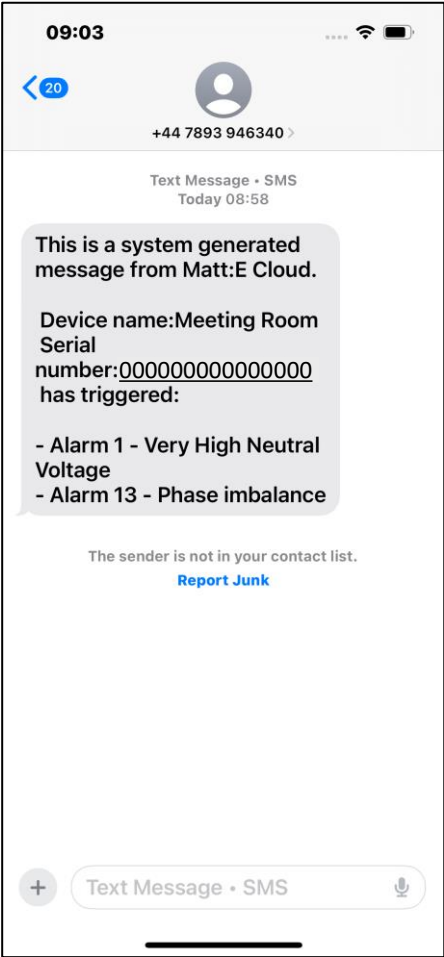
For the text alerts ensure to use the country code. +44 (UK) for example.



Click on the cog to enter the device settings page. Here you can enter the details for the contact in the event that a fault is detected.



EVENT NOTIFICATIONS



REMOTE STOP

Remote stop function is only toggled via the Portal - Device settings. This function is used to remotely Switch the NO/NC of both relays.

To Activate the remote stop, select “Activate Remote Stop” And confirm the change. (Both relays will return to the NC state)

To Deactivate the remote stop, select “Deactivate Remote Stop” And confirm the change. (Both relays will return to the NO state)

The notification of remote stop being activated shows on the Guardian device.

The confirmation is also shown on the General settings.

The screenshot shows the 'Device Settings' page with a modal dialog asking for confirmation to change the remote stop status. Below the dialog, there are input fields for 'Phone number including country code', 'Email 1', and 'Email 2', each with a corresponding 'Enable' checkbox. The 'Remote stop Status' section shows 'Remote stop is not activated - connected load is not isolated' and an 'Activate remote stop' button. A 'Close' button is at the bottom right.

This screenshot shows the 'Remote stop Status' section after activation. It displays 'Remote stop is activated - connected load is isolated' and a red 'Deactivate remote stop' button.

The screenshot shows a notification on the Guardian device titled 'Remote stop notification'. It states 'The remote stop has been: Activated' and prompts the user to 'Press "OK" to close notification' with an 'OK' button.

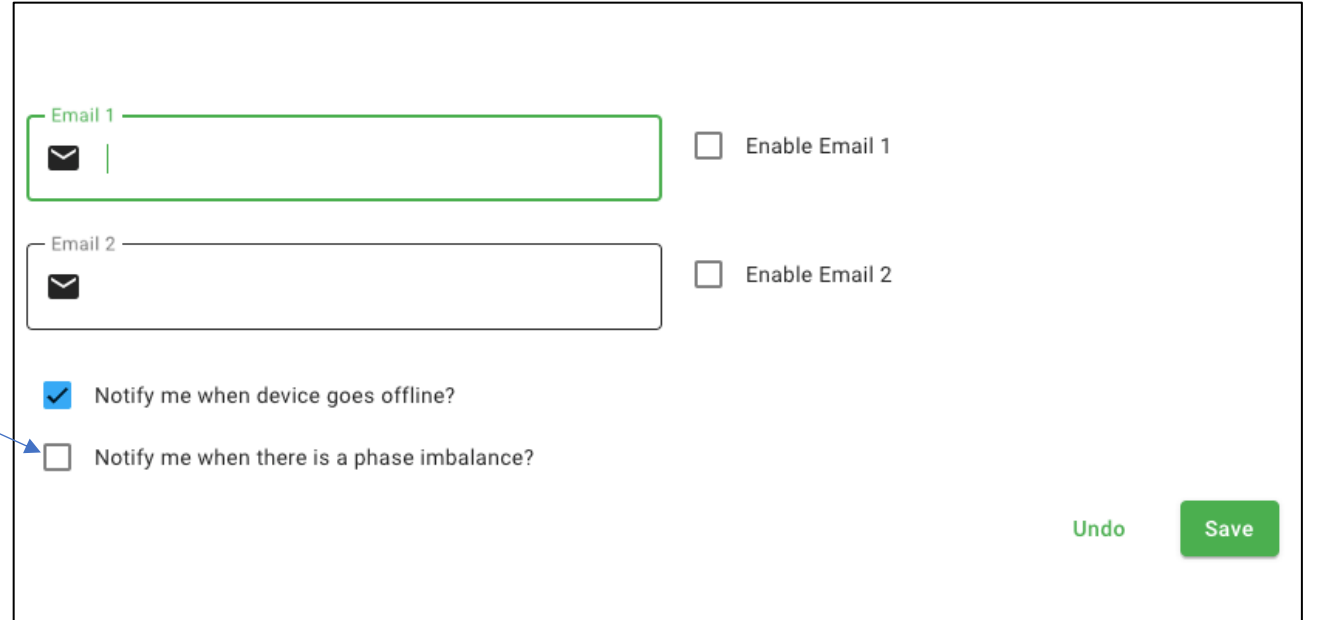
The screenshot shows the 'General settings' page. It lists various device information including 'Device name', 'Model name', 'Serial number', 'Firmware version', 'Bootloader version', 'UX revision', 'MAC address', 'Manufacture date', 'Claimed by', and 'Remote stop'. The 'Remote stop' status is listed as 'Not Active'.



PHASE IMBALANCE

It is possible to turn ON and OFF the notification for the phase imbalance.

Simply go to the unit device settings and toggle the notifications to suit.



The screenshot shows a settings interface for phase imbalance notifications. It features two email input fields, each with an envelope icon and a label 'Email 1' and 'Email 2' respectively. To the right of each email field is a checkbox labeled 'Enable Email 1' and 'Enable Email 2'. Below these are two checkboxes: 'Notify me when device goes offline?' (checked) and 'Notify me when there is a phase imbalance?' (unchecked). At the bottom right are 'Undo' and 'Save' buttons. A blue arrow points from the text 'toggle the notifications to suit.' to the 'Notify me when there is a phase imbalance?' checkbox.

Email 1 ☐ Enable Email 1

Email 2 ☐ Enable Email 2

☒ Notify me when device goes offline?

☐ Notify me when there is a phase imbalance?

Undo [Save](#)